

DevOps Lab

Documentation

Program 10:

Demonstrate minikube dashboard using CLI and Kubernetes Cluster

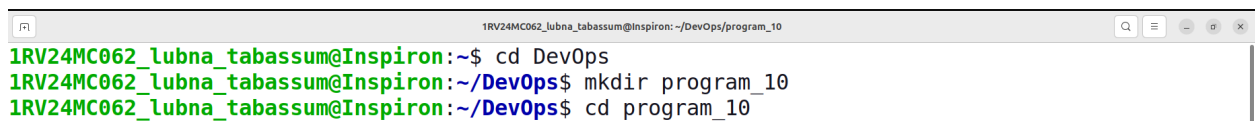
The program aims to demonstrate how to access and manage a Kubernetes cluster visually using the Minikube Dashboard through the Command Line Interface (CLI).

Project Structure:

```
program_10/  
—>deployment.yaml  
—>sesrvices.yaml
```

Procedure:

Create a folder (ex: program_10)

A terminal window screenshot showing the creation and navigation into a new directory. The terminal title bar reads '1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_10'. The command history shows: 1. 'cd DevOps' from the home directory. 2. 'mkdir program_10' from the DevOps directory. 3. 'cd program_10' from the DevOps directory.

```
1RV24MC062_lubna_tabassum@Inspiron:~$ cd DevOps  
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps$ mkdir program_10  
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps$ cd program_10
```

Step 1: Start Minikube

```
minikube start
```

Check the status of Minikube

```
minikube status
```

```

1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ minikube start
minikube v1.37.0 on Ubuntu 24.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Updating the running docker "minikube" container ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Enabled addons: storage-provisioner, default-storageclass

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

```

Step 2: Create deployment.yaml file with the following code
 nano deployment.yaml

```

GNU nano 7.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80

```

[Read 21 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Step 3: Create services.yaml file with the following code
 nano services.yaml

```
GNU nano 7.2 services.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort

[ Read 12 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Step 4: Apply the files

kubectl apply -f deployment.yaml

kubectl apply -f services.yaml

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ kubectl apply -f services.yaml
service/nginx-service created
```

Step 5: Get pods, deployment, services

kubectl get pods

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6xr9c  1/1     Running   0           12s
nginx-deployment-6f9664446b-qqs4    1/1     Running   0           12s
```

kubectl get deployment

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ kubectl get deployment
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 2/2      2             2           21s
```

kubectl get services

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ kubectl get services
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1       <none>       443/TCP          2d6h
nginx-service       NodePort    10.104.121.196  <none>       80:31868/TCP     24s
```

Step 6: Check the minikube services

minikube services nginx-service

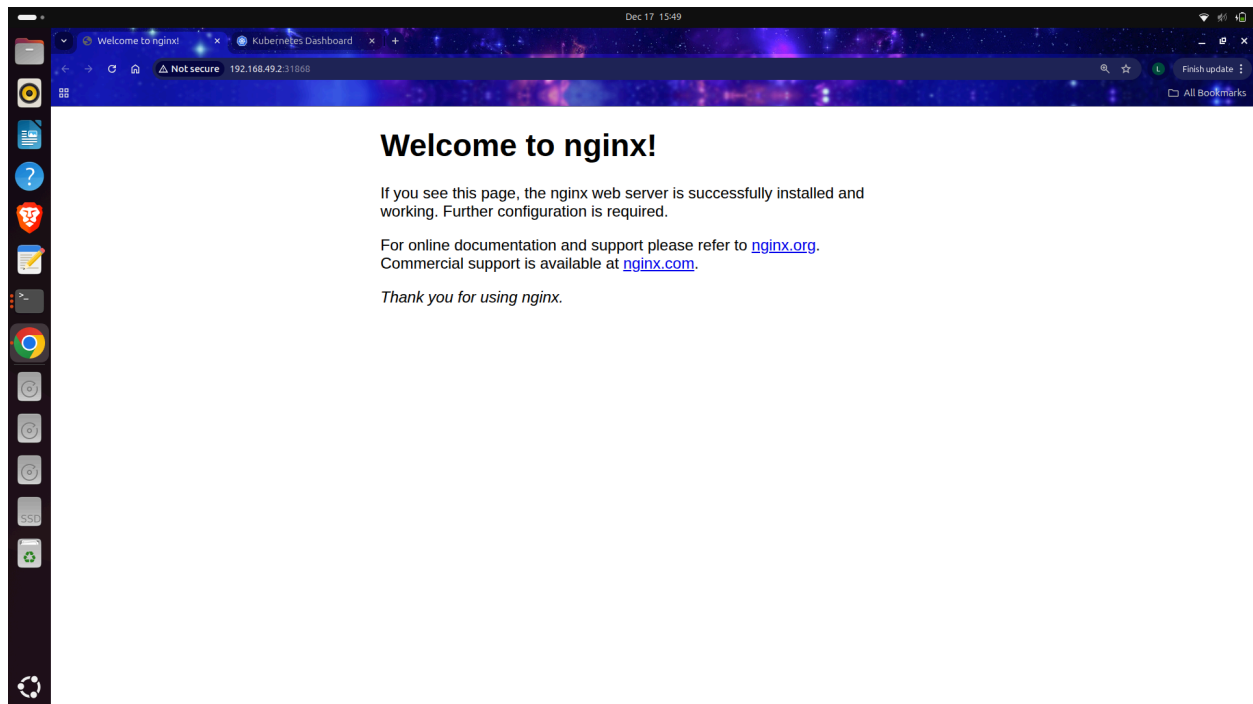
```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31868

🌐 Opening service default/nginx-service in default browser...

```
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_10$ WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1765966078.624379    61375 voice_transcription.cc:58] Registering VoiceTranscriptionCapability
Created TensorFlow Lite XNNPACK delegate for CPU.
Attempting to use a delegate that only supports static-sized tensors with a graph that has dynamic-sized tensors (tensor#-1 is a dynamic-sized tensor).
[61348:61378:1217/153759.645211:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[61348:61378:1217/153829.397816:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
[61348:61378:1217/153910.765424:ERROR:google_apis/gcm/engine/registration_request.cc:291] Registration response error message: DEPRECATED_ENDPOINT
```

Output:



Dec 17 15:49

Welcome to nginx!Kubernetes DashboardKubernetes Dashboard

127.0.0.1:41157/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/#/workloads?namespace=default

Finish updateAll Bookmarks

kubernetes

default

Search

+🔔

Workloads

Workloads

Cron Jobs

Daemon Sets

Deployments

Jobs

Pods

Replica Sets

Replication Controllers

Stateful Sets

Service

Ingresses

Ingress Classes

Services

Config and Storage

Config Maps

Persistent Volume Claims

Secrets

Storage Classes

Cluster

Cluster Role Bindings

Cluster Roles

Events

Workload Status

Running 1

Running 2

Running 1

Deployments

Pods

Replica Sets

Deployments

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	11 minutes ago

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
nginx-deployment-6f9664446b-6xr9c	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	11 minutes ago
nginx-deployment-6f9664446b-rvtf4	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	11 minutes ago